Анализ на времеви редове

AR(p) процеси. Приложение при моделиране на времеви редове с изразена цикличност.

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$AR(1)$ processes

$$X_t = c + \phi_1 X_{t-1} + \varepsilon_t$$

$$\varepsilon_t \sim i.i.d.(0, \sigma^2)$$

$$Y_t = X_t - \mu$$
$$Y_t = \phi_1 B Y_t + \varepsilon_t$$

$$\boldsymbol{(1 - \phi_1 B) Y_t = \varepsilon_t}$$

$AR(1)$ conditional moments

$$X_t = c + \phi_1 X_{t-1} + \varepsilon_t$$

$$E(X_t|X_{t-1}) = c + \phi_1 X_{t-1}$$

$$Var(X_t|X_{t-1}) = Var(\varepsilon_t) = \sigma^2$$

$AR(1)$ unconditional moments

$$E(X_t) = \mu$$

$$Var(X_t) = \gamma_0$$

$$cov(X_t, X_{t-j}) = \gamma_j$$

$AR(1)$ unconditional mean

1. $X_t = c + \phi_1 X_{t-1} + \varepsilon_t$

2. $E(X_t) = c + \phi_1 E(X_{t-1})$

3. $E(X_t) = E(X_{t-1}) = \mu$

4. $\mu = c + \phi_1 \mu$

5. $\mu = \dfrac{c}{1 - \phi_1}$

$AR(1)$ alternative representations

$$X_t - \mu = \phi_1(X_{t-1} - \mu) + \varepsilon_t$$
$$X_{t-1} - \mu = \phi_1(X_{t-2} - \mu) + \varepsilon_{t-1}$$

By repeated substitutions the latter equation implies that:

$$X_t - \mu = \varepsilon_t + \phi_1\varepsilon_{t-1} + \phi_1^2\varepsilon_{t-2} + \cdots =$$
$$= \sum_{i=0}^{\infty} \boldsymbol{\phi_1^i \, \varepsilon_{t-i}}$$

$AR(1)$ unconditional variance

$$1.\ X_t - \mu = \phi_1(X_{t-1} - \mu) + \varepsilon_t$$

$$2.\ (X_t - \mu)^2 = \phi_1^2(X_t - \mu)^2 + 2\phi_1(X_{t-1} - \mu)\varepsilon_t + \varepsilon_t^2$$

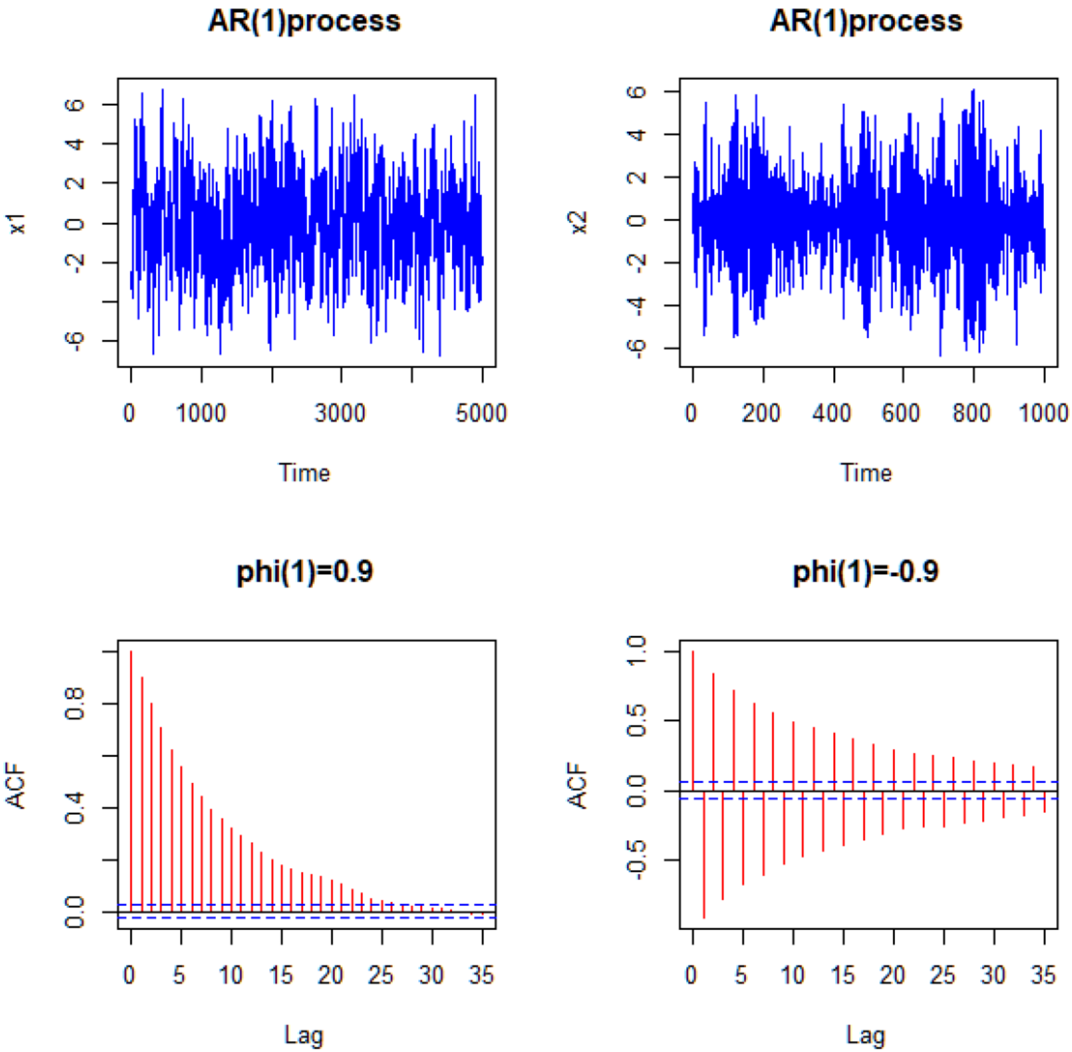
$$3.\ Var(X_t) = \phi_1^2 Var(X_{t-1}) + \sigma^2$$

$$4.\ Var(X_t) = \frac{\sigma^2}{1 - \phi_1^2}$$

$AR(1)$ stationarity condition

- The necessary and sufficient condition for the $AR(1)$ model to be weakly stationary is $|\boldsymbol{\phi_1}| < \mathbf{1}$.

ACF of $AR(1)$



$$\rho_l = \phi_1 \rho_{l-1}, l \geq 0$$

Because $\rho_0 = 1$, $\boldsymbol{\rho_l = \phi_1^l}$

$AR(2)$ processes

$$X_t = c + \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$$

$$\varepsilon_t \sim i.i.d.(0, \sigma^2)$$

$$Y_t = X_t - \mu$$
$$Y_t = \phi_1 B Y_t + \phi_2 B^2 Y_t + \varepsilon_t$$

$$\boldsymbol{(1 - \phi_1 B - \phi_2 B^2) Y_t = \varepsilon_t}$$

$AR(2)$ unconditional mean and ACF

$$\boldsymbol{E(X_t) = \mu = \frac{c}{1 - \phi_1 - \phi_2}, \phi_1 + \phi_2 \neq 1}$$

$$\boldsymbol{\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}, l > 0}$$

$$\rho_0 = 1$$

$$\rho_1 = \phi_1 \rho_0 + \phi_2 \rho_{-1} = \phi_1 + \phi_2 \rho_1$$

$$\rho_1 = \frac{\phi_1}{1 - \phi_2}$$

$$\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}, l \geq 2$$

$AR(2)$ ACF properties

$$\boldsymbol{\rho_l = \phi_1 \rho_{l-1} + \phi_2 \rho_{l-2}}$$

$$(1 - \phi_1 B - \phi_2 B^2)\rho_l = 0$$

$$1 - \phi_1 x - \phi_2 x^2 = 0$$

$$\omega_{1,2} = \frac{\phi_1 \pm \sqrt{\phi_1^2 + 4\phi_2}}{-2\phi_2}$$

$AR(2)$ ACF properties

$$(1 - \phi_1 B - \phi_2 B^2)\rho_l = 0$$
$$(1 - \omega_1 B)(1 - \omega_2 B)\rho_l = 0$$

If $\phi_1^2 + 4\phi_2 < 0$, then ω_1 and ω_2 are complex numbers.

Average length of the stochastic cycles

Let $\omega_{1,2} = a \pm bi$, $i = \sqrt{-1}$, then

$$k = \frac{2\pi}{\cos^{-1}\left(a/\sqrt{a^2 + b^2}\right)}$$

$AR(p)$ stationarity condition

- $1 - \phi_1 x - \phi_2 x^2 - \ \ldots - \phi_p x^p = 0$
- If all the solutions of the characteristic equation are greater than one in modulus, then the series is stationary.

References

Tsay, R., 2010. *Analysis of Financial Time Series.* 3rd ed. New Jersey: Wiley Series in Probability and Statistics, John Wiley & Sons.

Ch. 2.4.